

Design Pattern 11: Dependency Injection

Exercise 11: Implementing Dependency Injection

Scenario:

You are developing a customer management application where the service class depends on a repository class. Use Dependency Injection to manage these dependencies.

Steps:

Create a New Java Project:

Create a new Java project named `DependencyInjectionExample`.

Define Repository Interface:

Create an interface `CustomerRepository` with methods like `findCustomerById()`.

Implement Concrete Repository:

Create a class `CustomerRepositoryImpl` that implements `CustomerRepository`.

Define Service Class:

Create a class `CustomerService` that depends on `CustomerRepository`.

Implement Dependency Injection:

Use constructor injection to inject `CustomerRepository` into `CustomerService`.

Test the Dependency Injection Implementation:

Create a main class to demonstrate creating a `CustomerService` with `CustomerRepositoryImpl` and using it to find a customer.